

CHE 201 Spring 2020 Quiz 2

Anna Christensen

TOTAL POINTS

92 / 100

QUESTION 1

1 40 pts

1.1 1a 20 / 20

✓ - 0 pts Correct

- 5 pts not explained

☞ you do not need to convert into SI units. You can use A-11E table instead of A-11

1.2 1b 15 / 20

- 0 pts Correct

✓ - 5 pts v1 and v2 are incorrect

- 5 pts state 1 not shown

QUESTION 2

2 30 pts

2.1 2a 15 / 15

✓ - 0 pts Correct

- 2 pts calculation error

2.2 2b 15 / 15

✓ - 0 pts Correct

- 2 pts calculation error

QUESTION 3

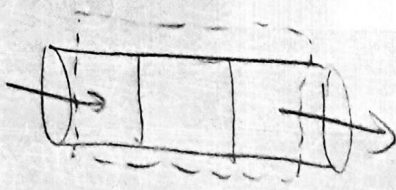
3 3 27 / 30

- 0 pts Correct

- 3 pts not explained

✓ - 3 pts calculation error, Q is -2kJ/s

①



$$P_2 < P_1$$

$$T_2 < T_1$$

open system
KE = PE = 0

$$T_1 = 115^\circ \text{F}$$

$$P_2 = 60 \frac{\text{lbf}}{\text{in}^2}$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} [(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2)]$$

$$0 = \dot{m} [(h_1 - h_2)]$$

$$h_1 = h_2$$

table A-10

saturated liquid

$$T = 115^\circ \text{F} = 46.1^\circ \text{C}$$

$$h @ 44^\circ = 112.22$$

between 44 and 48°C

$$@ 48^\circ = 118.35$$

$$P_2 =$$

$$60 \frac{\text{lbf}}{\text{in}^2} = 4.14 \text{ bar}$$

between 4 and 5 bar

$$\text{Finding } h \quad \frac{46.1 - 44}{48 - 44} = \frac{h_1 - 112.22}{118.35 - 112.22}$$

$$h_1 = 115.44 = h_2$$

$$\frac{4.14 - 4}{5 - 4} = \frac{T_2 - 8.93}{15.74 - 8.93}$$

$$T_2 = 9.88^\circ \text{C}$$

$$\text{Finding } P_1 \quad \frac{46.1 - 44}{48 - 44} = \frac{P_1 - 11.299}{12.526 - 11.299}$$

$$P_1 = 11.943 \text{ bar}$$

$$b) \quad \frac{46.1 - 44}{48 - 44} = \frac{v_1 - 0.8847}{0.8989 - 0.8847}$$

$$v_1 = 0.8922$$

$$v_2 =$$

$$h = (1-x)h_f + xh_g$$

$$115.44 = (1-x)h_f + xh_g$$

using table A-11

$$h_f = \frac{4.14 - 4}{5 - 4} = \frac{h_f - 62.0}{71.33 - 62.0}$$

$$h_f = 63.31$$

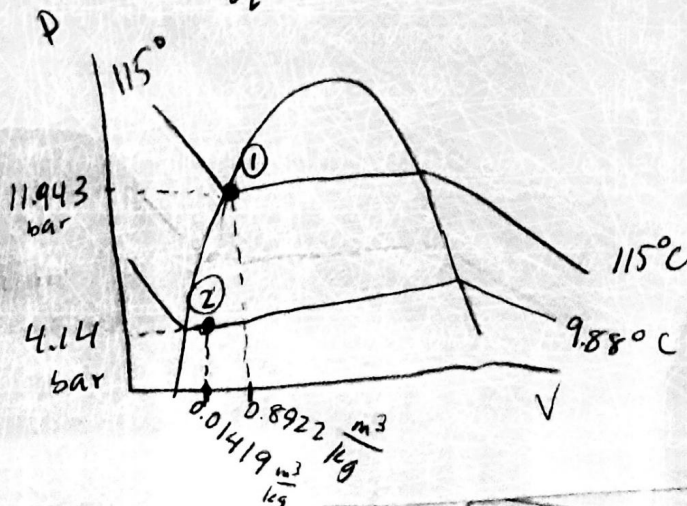
$$h_g = \frac{4.14 - 4}{5 - 4} = \frac{h_g - 252.32}{256.07 - 252.32}$$

$$h_g = 252.85$$

$$115.44 = (1-x)(63.31) + x(252.85)$$

$$115.44 = 63.31 - 63.31x + 252.85x$$

$$x = 0.275$$



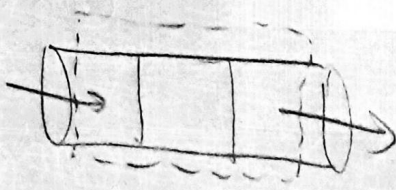
1.11a 20 / 20

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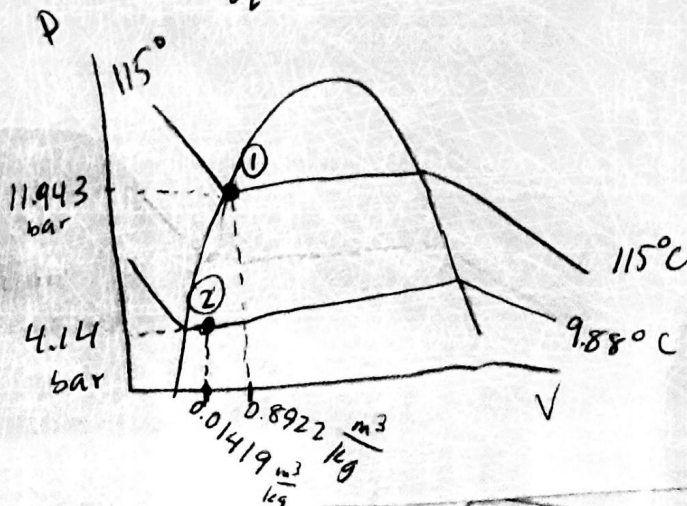
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$$115.44 = 63.31 - 63.31x + 252.85x$$

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(16)

$$v_{2f} = \frac{4.14 - 4}{5 - 4} = \frac{v_f - (0.7904 \times 10^{-3})}{(0.8056 - 0.7904) \times 10^{-3}}$$

$$v_f = 0.7925 \times 10^{-3}$$

$$v_{2g} = \frac{4.14 - 4}{5 - 4} = \frac{v_g - 0.0509}{0.0409 - 0.0509}$$

$$v_g = 0.0495$$

$$V_2 = (1-x)v_f + x v_g$$

$$V_2 = (1 - 0.275)(0.7925 \times 10^{-3}) + 0.275(0.0495)$$

$$V_2 = 0.01419 \frac{\text{m}^3}{\text{kg}}$$

1.2 1b 15 / 20

- 0 pts Correct

✓ - 5 pts v1 and v2 are incorrect

- 5 pts state 1 not shown

②

$$\dot{V} = 85 \frac{\text{L}}{\text{min}}$$

$$T_1 = 22^\circ \text{C}$$

$$P_1 = 3.6 \text{ bar}$$

$$d = 8 \text{ cm}$$

velocity = ?

$\dot{m} = ?$

$$a) \dot{V} = AV \quad A = \left(\frac{0.08 \text{ m}}{2} \right)^2 \pi = 0.00503 \text{ m}^2$$

$$85 \frac{\text{L}}{\text{min}} = 0.085 \frac{\text{m}^3}{\text{min}}$$

$$0.085 \frac{\text{m}^3}{\text{min}} = 0.00503 \text{ m}^2 \left(V \frac{\text{m}}{\text{min}} \right)$$

$$V = 16.9 \frac{\text{m}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} = \boxed{0.282 \frac{\text{m}}{\text{s}}}$$

ideal gas

$$b) \dot{m} = \frac{P \dot{V}}{RT} = \frac{(3.6 \text{ bar} \times 10^2 \frac{\text{kPa}}{\text{bar}}) \left(85 \frac{\text{L}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} \right)}{8.3145 \frac{\text{J}}{\text{K mol}} (22 + 273 \text{ K})} = 0.2079 \frac{\text{mol}}{\text{s}}$$

$$0.2079 \frac{\text{mol}}{\text{s}} \times \frac{0.02891 \text{ kg}}{\text{mol}} = \boxed{6 \times 10^{-3} \frac{\text{kg}}{\text{s}}}$$

2.1 2a 15 / 15

✓ - 0 pts Correct

- 2 pts calculation error

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2.2 2b 15 / 15

✓ - 0 pts Correct

- 2 pts calculation error

③

$$\dot{Q}_w = 2 \text{ kW}$$

$$P_1 = 1.05 \text{ bar}$$

$$T_1 = 300 \text{ K}$$

$$\dot{V} = AV = 12 \frac{\text{m}^3}{\text{min}}$$

$$P_2 = 12 \text{ bar}$$

$$T_2 = 400 \text{ K}$$

$$\dot{W}_{cv} = ?$$

• open system

• ideal gas

• KE = PE = 0

• one input, one output

$$0 = \dot{Q}_w - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2) \right] \Rightarrow \dot{W}_{cv} = \dot{Q}_w + \dot{m}(h_1 - h_2)$$

$$\dot{m} = \frac{P_1 \dot{V}}{RT_1} = \frac{(1.05 \text{ bar} \times 10^5 \frac{\text{Pa}}{\text{bar}}) (12 \frac{\text{m}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}})}{8.3145 \frac{\text{m}^3 \text{ Pa}}{\text{K mol}} (300 \text{ K})} = 8.419 \frac{\text{mol}}{\text{s}} \times \frac{0.028 \text{ kg}}{1 \text{ mol}}$$

table A-22

$$h_1 = 300.19 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 = 400.98 \frac{\text{kJ}}{\text{kg}}$$

$$= 0.243 \frac{\text{kg}}{\text{s}}$$

$$\begin{aligned} \dot{W}_{cv} &= 2 \text{ kW} + 0.243 \frac{\text{kg}}{\text{s}} \left(300.19 \frac{\text{kJ}}{\text{kg}} - 400.98 \frac{\text{kJ}}{\text{kg}} \right) \\ &= 2 \text{ kW} + -24.5 \frac{\text{kJ}}{\text{s}} = \boxed{-22.5 \text{ kW}} \end{aligned}$$

33 27 / 30

- 0 pts Correct

- 3 pts not explained

✓ - 3 pts calculation error, Q is -2kj/s